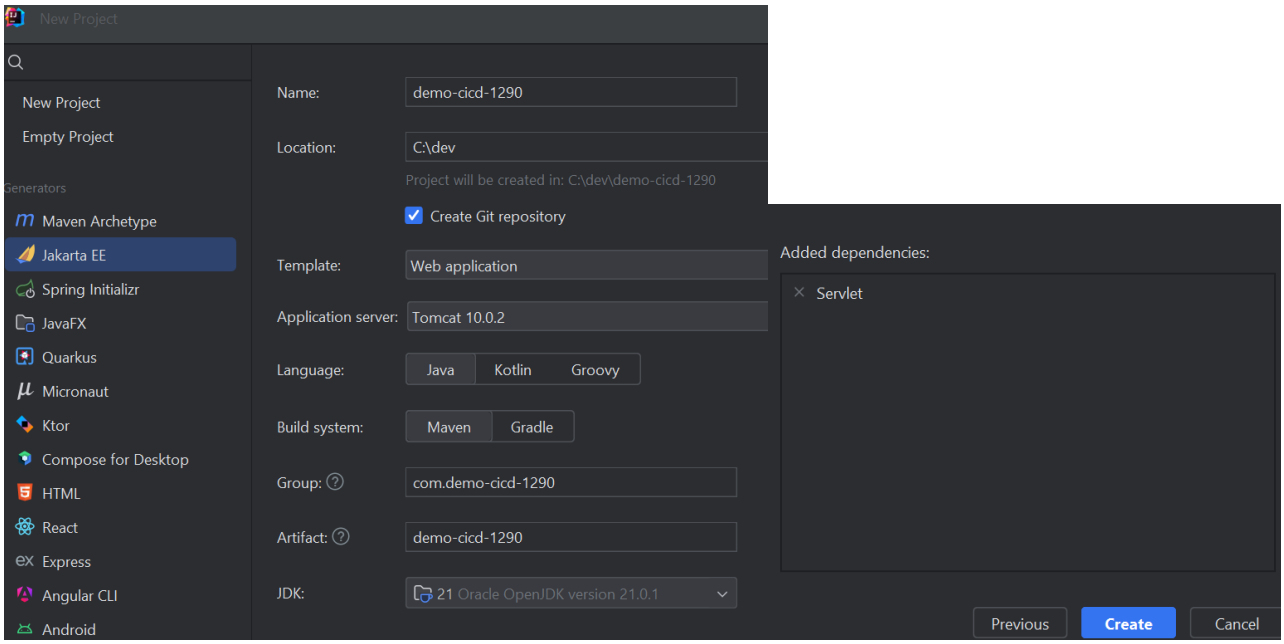
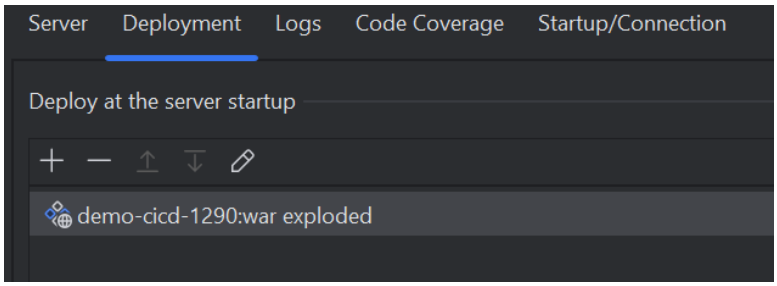
Atelier pratique : Jenkins

Création d'un pipeline CICD

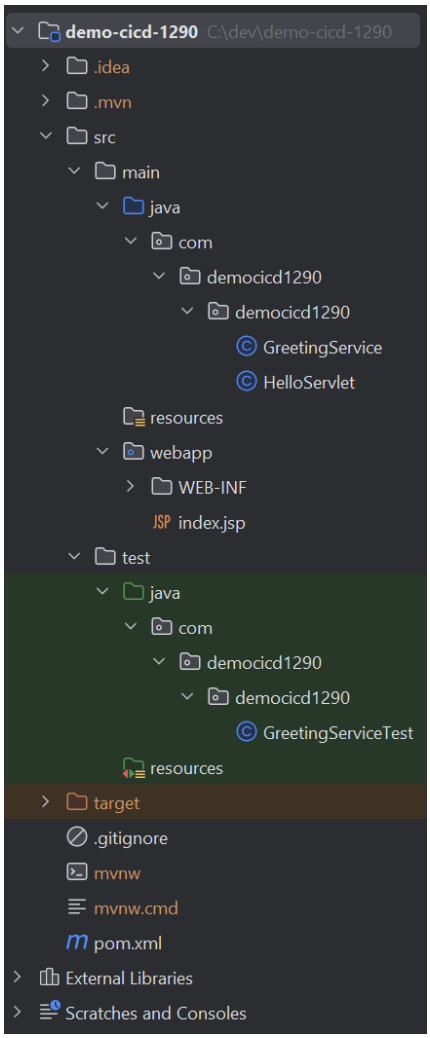
a- Création d'un projet JakartaEE :

- Java 21 via maven.compiler.release
- Jakarta Servlet API en provided
- JaCoCo : rapport (ex. 60%)
- Surefire JUnit 5

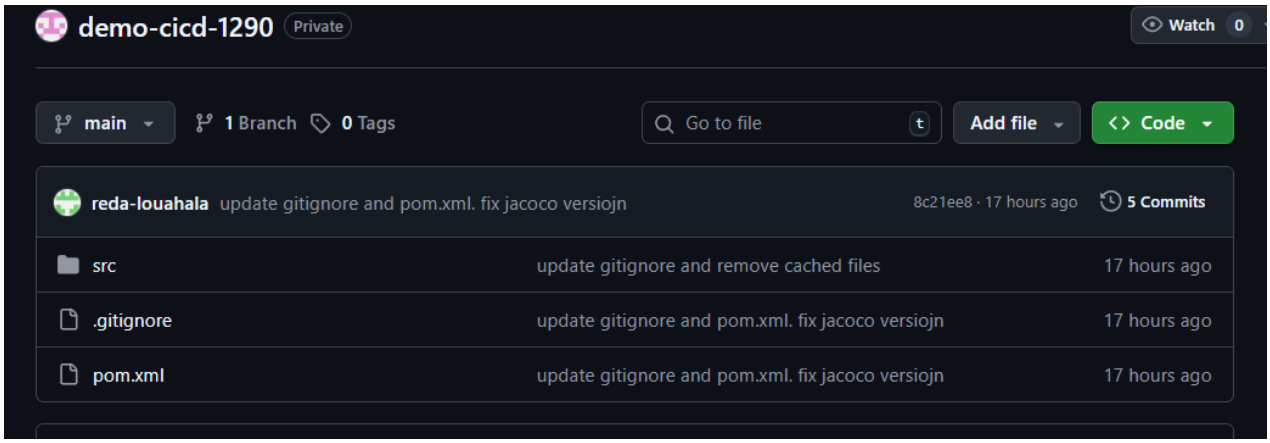




Voici à quoi ressemble la structure du projet :



b- Création du repo privé GitHub et le synchroniser avec le projet crée :



- Créez un jeton pour votre repo GitHub

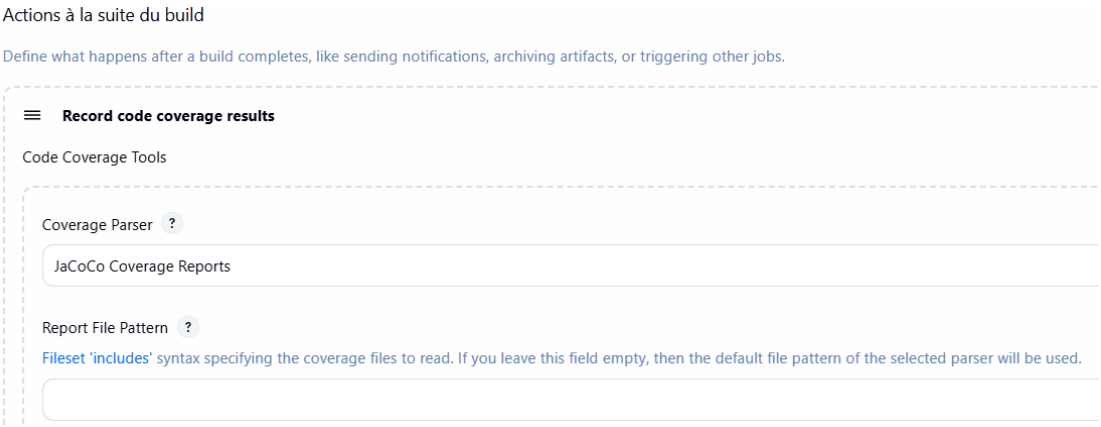
c- Installer Jenkins avec jdk 21 (si nécessaire), exemple sous docker:

docker run -d -p 8091:8080 -p 50000:50000 -v jenkins_home:/var/jenkins_home --name jenkins-1290 jenkins/jenkins:lts-jdk21

d- Créez et complétez la configuration de votre pipeline sur Jenkins (git, github, credentials, etc.)

e- Ajoutez le plugin JaCoCo pour générer le rapport de couverture des tests :





f- Exécutez et valider le bon fonctionnement de votre pipeline

```
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.democicd1290.democicd1290.GreetingServiceTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.081 s -- in com.democicd1290.democicd1290.GreetingServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.10:report (report) @ demo-cicd-1290 ---
[INFO] Loading execution data file /var/jenkins_home/workspace/demo-1290_test/target/jacoco.exec
[INFO] Analyzed bundle 'demo-cicd-1290' with 2 classes
```

g- Faites un changement au niveau des tests pour forcer un échec, et réexécuter le pipeline

```
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.democicd1290.democicd1290.GreetingServiceTest
[ERROR] Tests run: 3, Failures: 1, Errors: 0, Skipped: 0, Time elapsed: 0.101 s <<< FAILURE! -- in com.democicd1290.democicd1290.GreetingServiceTest
[ERROR] com.democicd1290.democicd1290.GreetingServiceTest.isQuietHours_trueAtNight_falseAtDay -- Time elapsed: 0.007 s <<< FAILURE!
org.opentest4j.AssertionFailedError: expected: <true> but was: <false>
        at org.junit.jupiter.api.AssertionFailureBuilder.build(AssertionFailureBuilder.java:151)
        at org.junit.jupiter.api.AssertionFailureBuilder.buildAndThrow(AssertionFailureBuilder.java:132)
        at org.junit.jupiter.api.AssertTrue.failNotTrue(AssertTrue.java:63)
        at org.junit.jupiter.api.AssertTrue.assertTrue(AssertTrue.java:36)
        at org.junit.jupiter.api.AssertTrue.assertTrue(AssertTrue.java:31)
        at org.junit.jupiter.api.Assertions.assertTrue(Assertions.java:180)
```

Rapport de test :

Dans IntelliJ :

demo-cicd-1290

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
com.democicd1290.democicd1290		59 %		83 %	6	14	10	19	4	8	1	2
Total	33 of 82	59 %	2 of 12	83 %	6	14	10	19	4	8	1	2

com.democicd1290.democicd1290

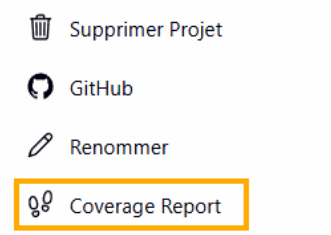
Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
HelloServlet		0 %		n/a	4	4	10	10	4	4	1	1
GreetingService		87 %		83 %	2	10	0	9	0	4	0	1
Total	33 of 82	59 %	2 of 12	83 %	6	14	10	19	4	8	1	2

Etc.

Dans Jenkins :

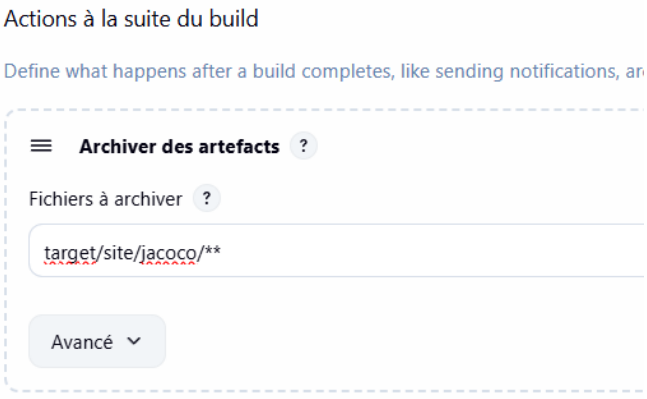
```
[Coverage] Recording coverage results
[Coverage] Creating parser for JaCoCo Coverage Reports
[Coverage] Using default pattern '**/jacoco.xml,**/jacocoTestReport.xml' since user defined pattern is not set
[Coverage] Searching for all files in '/var/jenkins_home/workspace/demo-1290_test' that match the pattern '**/jacoco.xml,**/jacocoTestReport.xml'
[Coverage] Traversing of symbolic links: enabled
[Coverage] -> found 1 file
[Coverage] Successfully parsed file '/var/jenkins_home/workspace/demo-1290_test/target/site/jacoco/jacoco.xml'
[Coverage] MODULE: 100.00% (1/1)
[Coverage] PACKAGE: 100.00% (1/1)
[Coverage] FILE: 50.00% (1/2)
[Coverage] CLASS: 50.00% (1/2)
[Coverage] METHOD: 50.00% (4/8)
[Coverage] LINE: 47.37% (9/19)
[Coverage] BRANCH: 83.33% (10/12)
[Coverage] INSTRUCTION: 59.76% (49/82)
```

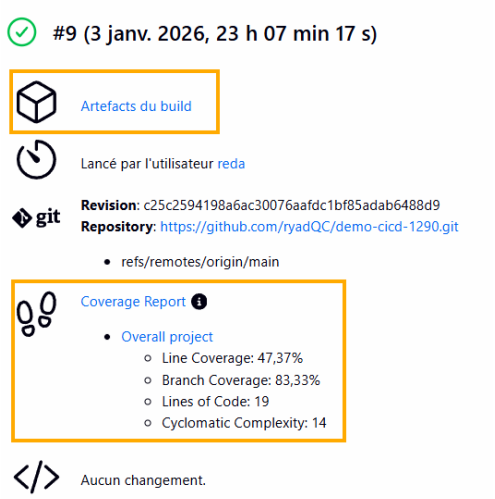
Ou bien :





h- On peut aussi ajouter une action qui permet de sauvegarder l'artefact (rapport) :

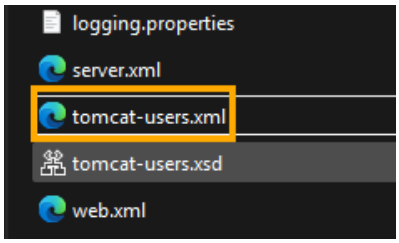






i- Démarrez Tomcat :

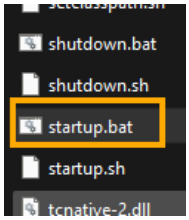
- Il faut d'abord faire une configuration des rôles :

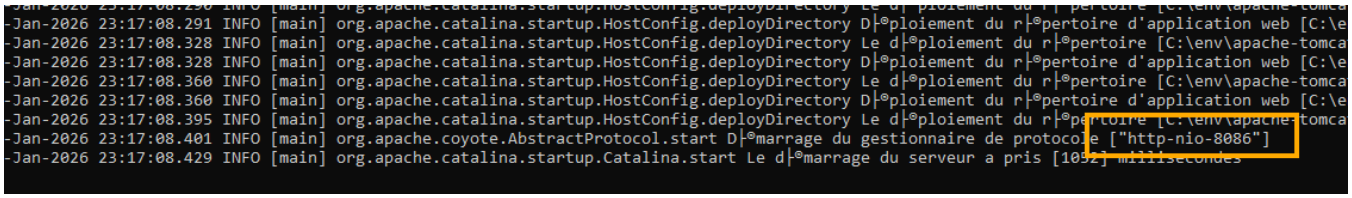



```
<role rolename="tomcat"/>
<role rolename="admin-gui"/>
<role rolename="manager-gui"/>
<role rolename="manager-script"/>

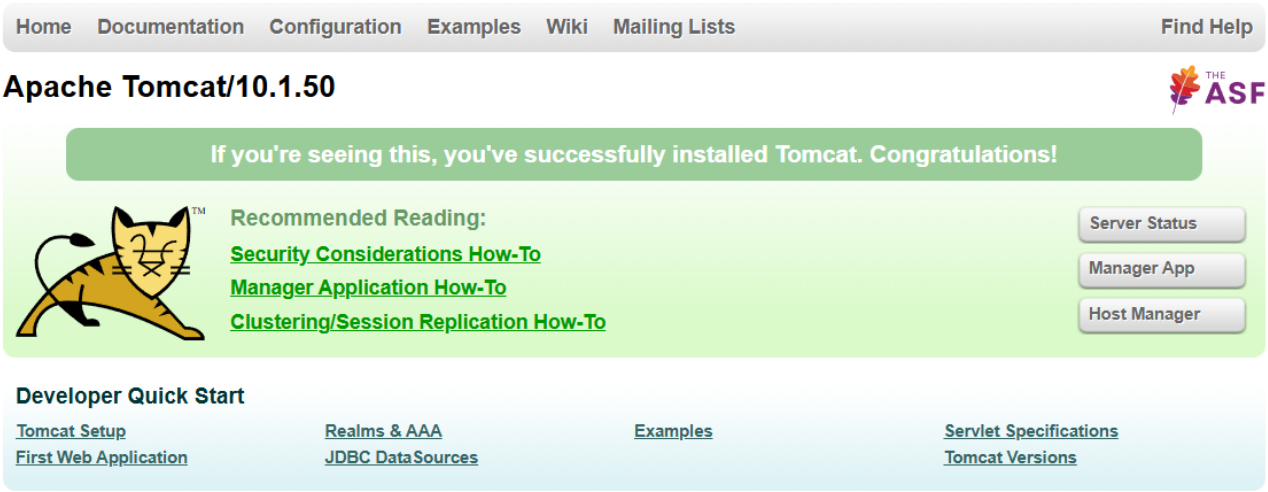
<user username="admin" password="admin" roles="admin-gui,manager-gui,manager-script"/>
```

- Dans le répertoire bin :

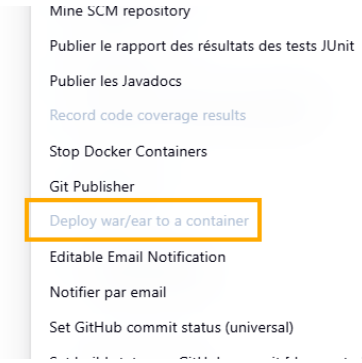


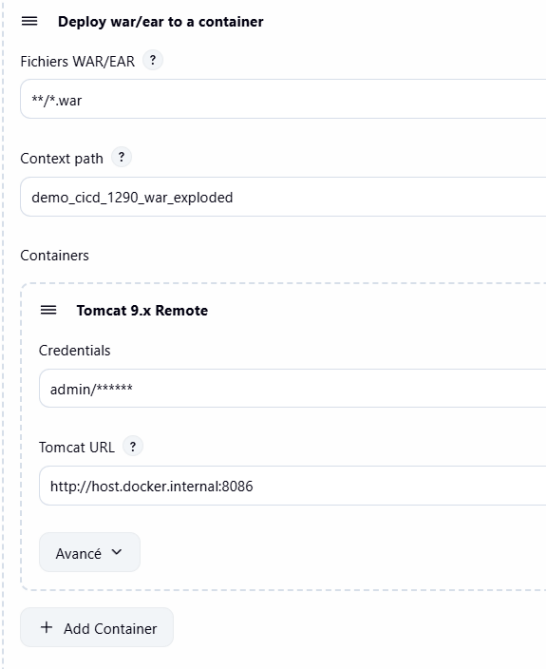


- Accédez via localhost :numPort pour faire un petit tour 😊



j- Ajouter une étape après la construction pour déployer dans Tomcat :






```
[Coverage] -> done
[Coverage] Deleting source code files of build #11
[Coverage] Finished coverage processing - adding the action to the build...
[DeployPublisher][INFO] Attempting to deploy 1 war file(s)
[DeployPublisher][INFO] Deploying /var/jenkins_home/workspace/demo-1290_test/target/demo-cicd-1290-1.0-SNAPSHOT.war to conta
  [/var/jenkins_home/workspace/demo-1290_test/target/demo-cicd-1290-1.0-SNAPSHOT.war] is not deployed. Doing a fresh deploym
  Deploying [/var/jenkins_home/workspace/demo-1290_test/target/demo-cicd-1290-1.0-SNAPSHOT.war]
Finished: SUCCESS
```

k- Vérifiez que l'application est déployée et quelle est en cours d'exécution :

Dans Tomcat :

Applications	
Chemin	**Version**
/	*Aucun spécifié*
/demo_cicd_1290_war_exploded	*Aucun spécifié*
/docs	*Aucun spécifié*

A partir d'un navigateur web :



Exercice :

Complétez l'automatisation du pipeline.